MIL-C-43062D
14 March 1988
SUPERSEDING
MIL-C-43062C
24 June 1982

MILITARY SPECIFICATION

CLOTH, COATED, NYLON, RESIN MODIFIED BUTYL COATED, ACID AND FUEL RESISTANT

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document is for one type of modified butyl rubber coated cloth, in the unvulcanized state. It is to be used for the manufacture of acid and fuel resistant clothing and is vulcanized after assembly of the clothing (see 6.1).

2. APPLICABLE DOCUMENTS

* 2.1 Government documents.

2.1.1 Documents. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

O-C-105 - Calcium Chloride, Dihydrate and Calcium Chloride, Anhydrous; Technical

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Natick Research, Development and Engineering Center, Natick, MA 01760-5014, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 8305

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T20:28Z
Check the source to verify that this is the current version before use.

MILITARY

MIL-T-5624	-	Turbine Fuel, Aviation, Grades JP-4 and JP-5
MIL-P-7254	-	Propellants, Nitric Acid
MIL-C-19377	-	Cloth, Twill, Nylon
MIL-P-51493	-	Paper, Chemical Agent Detector: M9

STANDARDS

FEDERAL

FED-STD-191	-	Textile Test Methods
FED-STD-601	-	Rubber; Sampling and Testing

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes

(Copies of documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.1.2 Other Government documents. The following other Government documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

JOINT COMMITTEE OF PRINTING (JCP) PUBLICATIONS

Government Paper Specification Standards JCP F10-Manifold, White and Colored

(Application for copies may be obtained from the Superintendent of Documents, US Government Printing Office, Washington, DC 20402-0001.)

* 2.2 Other publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issues of the nongovernment documents which are current on the date of the solicitation.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T20:28Z
Check the source to verify that this is the current version before use.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 618 - Conditioning Plastics and Electrical Insulating Material for Testing

D 1424 - Tear Resistance of Woven Fabrics by Falling-Pendulum (Elmendorf) Apparatus

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1187.)

(Technical society and technical association documents are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document shall take precedence. Nothing in this document, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Laboratory report approval. Unless otherwise specified (see 6.2), at the time of submission of a bid the contractor shall submit to the contracting officer a certified copy of a recent laboratory report covering material which he proposes to deliver. Unless otherwise specified by the contracting officer, the contractor shall certify that the material was manufactured in a plant where the coating will be performed if a contract is awarded. This laboratory report shall contain test data which demonstrates that the finished product proposed by the contractor has been tested in conformance with and found to comply with the requirements of this document. Any of the following types of reports will be satisfactory from the standpoint of this requirement:

a. An independent or commercial laboratory report.
b. The prospective contractor's own laboratory report.
c. A governmental laboratory report from a contract within two years of date of submission of bid.

The purpose of the above requirements is to assist the Government in determining the capability of bidders to manufacture a cloth meeting all the requirements of this document. The submission of an acceptable report under this requirement shall not be construed as relieving a contractor from subsequently meeting all requirements of the document on all deliveries.

3.2 Material.

3.2.1 Base cloth. The base cloth shall be nylon twill conforming to MIL-C-19377, except that the cloth shall be undyed.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T20:28Z
Check the source to verify that this is the current version before use.

3.2.2 Coating compound. The compound used for coating the base cloth shall be a composition of resin modified butyl rubber, pigmented to produce a black color and formulated in accordance with the requirements of this document (see 4.3.1.1).

* 3.2.3 Dusting powder. The dusting powder for application to the coated cloth shall be whiting, talc, or other finely divided mineral material that does not support mildew growth (see 4.3.1.1). Talc shall not contain asbestos and whiting should contain less than 1 percent quartz.

3.3 Coated cloth.

3.3.1 Application of coating. The base cloth shall be coated on both sides with the coating compound specified in 3.2.2. Approximately 25 percent of the coating shall be applied to one side of the cloth, and the balance shall be applied to the other side (see 4.3.1.1). After coating, the cloth shall be dusted on both sides with the powder specified in 3.2.3 to produce a uniform dull finish.

* 3.3.2 Physical requirements of vulcanized coated cloth. The physical properties of the coated cloth when vulcanized shall conform to the requirements specified in table I and in 3.3.2.1 through 3.3.2.5 when tested as specified in 4.3.2.

TABLE I. Physical requirements of vulcanized coated cloth

Characteristic	Requirement	
	Minimum	Maximum
Weight, oz/sq yd	10.0	13
Breaking strength, lbs		
Warp	80	--
Filling	80	--
Tearing strength, grams		
Warp	960	--
Filling	960	--
Hydrostatic resistance, lbs/sq in		
Initial	80	--
After strength of coating	60	--
After cold crack resistance	60	--

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T20:28Z
Check the source to verify that this is the current version before use.

TABLE I. Physical requirements of vulcanized coated cloth - Continued

Characteristic	Requirement Minimum	Maximum
Adhesion of coating, lbs/2-inch width	7.0	--
Stiffness, gms, warp only		
At 70°F	--	9.0
At - 20°F	--	12.0
Blocking, scale rating	(1)	
Nitric acid surface tack, rating	II	

3.3.2.1 Abrasion resistance. No visible loose fibers of the base cloth shall be exposed in the center 1 inch of the abraded portion of any of the specimens when tested as specified in 4.3.2. Visible loose fibers shall be regarded as fibers from the nylon yarns used to weave the base cloth and not as entire yarns or yarn plies.

3.3.2.2 Resistance to accelerated weathering. The test specimen shall not become soft and tacky nor stiff and brittle when compared to an unexposed specimen of the same sample unit when tested as specified in 4.3.2.

3.3.2.3 Fuel resistance. There shall be no evidence of cracking, stiffening, flaking, or separation of the coating from the base cloth when tested as specified in 4.3.2.

3.3.2.4 Acid resistance. There shall be no evidence of cracking, stiffening, flaking, separation of the coating from the base cloth or change in color of the indicator paper when tested as specified in 4.3.2.

3.3.2.5 Odor. The vulcanized coated cloth shall be free of objectionable odor when tested as specified in 4.3.2. Odors normally attributable to modified butyl rubber shall not be regarded as objectionable.

* 3.4 Toxicity clearance. Only those rubber formulations already approved by the appropriate medical service for contact with human skin and so listed in the invitation for bids or request for proposal shall be considered acceptable for the related procurements (see 6.3).

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T20:28Z
Check the source to verify that this is the current version before use.

3.5 Workmanship. The finished cloth shall conform to the quality of product established by this document and the occurrence of defects shall not exceed the applicable acceptable quality levels.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this document where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this document shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirement in the document shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Certificate of compliance. When certificates of compliance are submitted, the Government reserves the right to inspect such items to determine the validity of the certification.

4.3 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.3.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this document or applicable purchase document.

* 4.3.1.1 Certification. Components, materials, and processes listed below shall be accepted on the basis of a contractor's certificate of compliance with the indicated requirements:

a. The coating compound is a composition of resin modified butyl rubber, black in color.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T20:28Z
Check the source to verify that this is the current version before use.

b. The dusting powder is a mineral material that does not support mildew growth and contains no asbestos and less than 1 percent quartz.

c. The distribution of coating requirement has been met.

4.3.2 End item testing. The coated cloth shall be tested for the characteristics listed in table II. The methods of testing specified in FED-STD-191, whenever applicable and as listed in table II, shall be followed. The physical and chemical values specified in Section 3 apply to the average of the determinations made on a sample unit for test purposes as specified in the applicable test methods. The sample unit for testing shall be 3 continuous yards full width of the cloth vulcanized in the same manner in which the assembled garments are to be vulcanized, with respect to temperature, curing time and any other controlled processing factor. The lot size shall be expressed in units of 1 linear yard and the sample size (number of sample units) shall be as specified below. The lot shall be unacceptable if any sample unit fails to meet any requirement specified. All test reports shall contain the individual values utilized in expressing the final result.

Lot size (yards)	Sample size
800 or less	2
801 up to and including 22,000	3
22,001 and over	5

* 4.3.2.1 Standard test conditions. Test results obtained under testing conditions defined in FED-STD-191, ASTM D 618 (Procedure A), or FED-STD-601 will be acceptable except in case of dispute. In dispute cases, tests shall be conducted with both the specimen and test apparatus under standard conditions as defined in FED-STD-191.

TABLE II. End item testing

Characteristic	Requirement reference	Test method
Weight, overall	3.3.2	5041
Breaking strength	3.3.2	5100
Tearing strength	3.3.2	ASTM D 1424

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T20:28Z
Check the source to verify that this is the current version before use.

TABLE II. End item testing - Continued

Characteristic	Requirement reference	Test method
Hydrostatic resistance:		
Initial	3.3.2	5512
After strength of coating	3.3.2	5972 and 5512
After cold crack	3.3.2	5874 1/ and 5512
Adhesion of coating	3.3.2	5970
Stiffness, (warp only):		
At 70°F	3.3.2	5204 2/
At -20°F	3.3.2	5204 2/ 3/
Blocking	3.3.2	5872
Nitric acid surface tack	3.3.2	4.4.1.5
Abrasion resistance	3.3.2.1	5304 4/
Resistance to accelerated weathering	3.3.2.2	5804 5/
Chemical resistance:		
Fuel resistance	3.3.2.3	4.4.1.3
Acid resistance	3.3.2.4	4.4.1.4
Odor	3.3.2.5	4.4.2

1/ The specimens shall be conditioned at -40° ± 2°F for a minimum of 4 hours.

2/ The test shall be conducted in still air.

3/ The specimens and the instrument shall be conditioned at -20° ± 2°F for a minimum of 4 hours.

4/ The heavily coated side of the cloth shall be subjected to 40 cycles under a tension of 6 pounds, and under a load of 3 pounds pressure on the test abrader. The abradant shall be "fine" emery cloth. Only three specimens in the warpwise direction shall be tested, and the abrasion resistance shall be determined by visual examination.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T20:28Z
Check the source to verify that this is the current version before use.

5/ The specimen, with the heavily coated side of the cloth facing the light source, shall be exposed in the weatherometer for 100 hours.

4.4 Methods of inspection.

4.4.1 Chemical resistance.

* 4.4.1.1 Apparatus and chemicals.

a. Test jig No. 2 made of aluminum 1100 (see figure 1).

b. Wax - The wax shall be paraffin or a mixture of paraffin wax and beeswax containing not more than one-third beeswax.

c. Petroleum jelly.

d. Aluminum block - Dimensions approximately 4 inches by 2-7/16 inches by 1/2 inch.

e. Liquid vesicant detector paper conforming to MIL-P-51493.

f. Propellant nitric acid conforming to type III of MIL-P-7254.

g. Glass supporting and cover plates measuring 8 inches by 8 inches.

h. Distilled water.

i. Grade JP-4 fuel conforming to MIL-T-5624.

j. Onion skin paper conforming to JCP F10 (see 2.1).

k. Roller - Approximately 5-1/2 inches in diameter, 2 inches wide, and 10 pounds in weight.

l. Desiccator.

m. Calcium chloride, anhydrous, conforming to type II, grade A of O-C-105.

4.4.1.2 Preparation of test specimen. The test specimen shall be a section of the coated cloth 6 inches in diameter and shall be prepared as follows:

a. Place the test specimen between the spacer inserts and seal with melted wax. The wax shall form a continuous bead covering the outer circumference of the insert assembly.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T20:28Z
Check the source to verify that this is the current version before use.

b. Place the insert assembly in the test jig sleeve and fill the cavity between the assembly and sleeve wall with melted wax. Mate jig sleeve and jig collar and screw hand tight.

c. Seal the inside of the sleeve collar with melted wax. The wax seal shall be continuous and shall extend around and out onto the test specimen approximately 1/8 inch above the top of the spacer insert. Turn the jig assembly over and seal the bottom cavity with melted wax in similar fashion.

d. Place test assembly in desiccator over the calcium chloride and maintain for a minimum of 16 hours.

4.4.1.3 Fuel resistance. The test specimen shall be prepared as specified in 4.4.1.2 using the apparatus and chemicals specified in 4.4.1.1. Immediately after removal of the assembly from the desiccator, the test specimen shall be covered with 50 mL of JP-4 fuel for 15 minutes at standard conditions. After fuel exposure, the test specimen shall be removed from the test jig, rinsed in distilled water, dried for approximately 4 hours at standard conditions, and examined for requirements in 3.3.2.3.

4.4.1.4 Acid resistance. The test for acid resistance shall be performed as follows:

a. Prepare the test specimen as specified in 4.4.1.2 using the apparatus and chemicals specified in 4.4.1.1.

b. After conditioning, remove the test assembly from the desiccator and immediately perform the acid exposure test as follows:

1. Coat bottom and top edges of the test jig with 1/32 inch of petroleum jelly.

2. Place aluminum block in the center of a glass supporting plate with a piece of the vesicant paper, coated side up, on top of the block. Place the test assembly over the block with the heavily coated side of the test specimen up.

3. Cover the test specimen with 50 mL of propellant nitric acid and cover the test assembly with a glass cover plate.

c. After 2 hours acid exposure at standard conditions, remove the test specimen from the test jig, rinse in distilled water, dry at standard conditions for approximately 4 hours, and examine for requirements specified in 3.3.2.4.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T20:28Z
Check the source to verify that this is the current version before use.

4.4.1.5 Nitric acid surface tack. The nitric acid surface tack test shall be performed as follows:

a. The test specimen shall be prepared as specified in 4.4.1.2 using the apparatus and chemicals specified in 4.4.1.1.

b. Immediately after removal of the assembly from the desiccator, the test specimen shall be covered with 50 mL of propellant nitric acid for 15 minutes at standard conditions.

c. After acid exposure, the specimen shall be removed from the test jig, rinsed in distilled water, and dried for approximately 4 hours at standard conditions.

d. The wax lines shall be removed from the specimen by scraping or trimming. The specimen shall then be placed, exposed side up, on a smooth hard surface and covered with a 2 inch by 4 inch piece of onion skin paper.

e. The roller shall be passed over the full length of the paper covered test specimen twice in each direction (four passes) with pressure applied by the weight of the roll only.

f. After rolling, the onion skin paper shall be peeled from the test specimen and the surface tack determined by comparison with figure 2 and rating standards as specified in 4.4.1.5.1.

4.4.1.5.1 Rating standards for surface tack test (see figure 2 for pictorial representation of rating standard).

Rating I - No trace 1/ of coating material on the test paper nor any trace of paper fiber adhering to surface of the test specimen.

Rating II - A trace 1/ of coating material retained on the test paper or a trace of paper fiber adhering to the test specimen surface.

Rating III - Any degree of coating material or paper fiber transfer in excess of rating II.

1/ A trace is defined as coating material or paper fiber transfer covering 25 percent of the test specimen area.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T20:28Z
Check the source to verify that this is the current version before use.

4.4.2 Odor. The examination for objectionable odors shall be made on one specimen, 4 inches by 6 inches minimum, that has been freely exposed to circulating air at standard conditions for 24 hours ± 1 hour. The specimen shall fail if it displays odors other than those normally attributable to modified butyl rubber.

5. PACKAGING

This section is not applicable to this document.

6. NOTES

6.1 Intended use. The coated cloth is intended for the manufacture of protective clothing which is resistant to rocket fuels and oxidizers and suitable for use at low temperatures. The protective clothing will be assembled using unvulcanized coated cloth and vulcanized after assembly of the clothing.

6.2 Ordering data. Acquisition documents for the protective clothing should specify the following:

a. Title, number, and date of this document.
b. When laboratory report approval is not required (see 3.1).

6.3 Rubber formulations. Approval of such rubber formulations are the responsibility of the U.S. Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5014, and is based on more extensive tests, including those for toxicity, which are not set forth in this document. Because of the time necessary to conduct full evaluation, only those formulations already approved and so listed in the invitation for bids or request for proposal shall be considered acceptable for the related procurement (see 3.4).

* 6.4 Subject term (key word) listing.

Acid resistant
Butyl coated
Coated cloth
Fuel resistant
Nylon
Protective clothing

6.5 Changes from previous issue. The margins of this document are marked with an asterisk (*) to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only, and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content, as written, irrespective of the marginal notations and relationship to the last previous issue.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T20:28Z
Check the source to verify that this is the current version before use.

Custodians:

Army - GL
Navy - NU
Air Force - 99

Review activities:

Army - MD, ME
Navy - SH
Air Force - 82
DLA - CT

User activity:

Air Force - 45

Preparing activity:

Army - GL

Project No. 8305-0184

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T20:28Z
Check the source to verify that this is the current version before use.

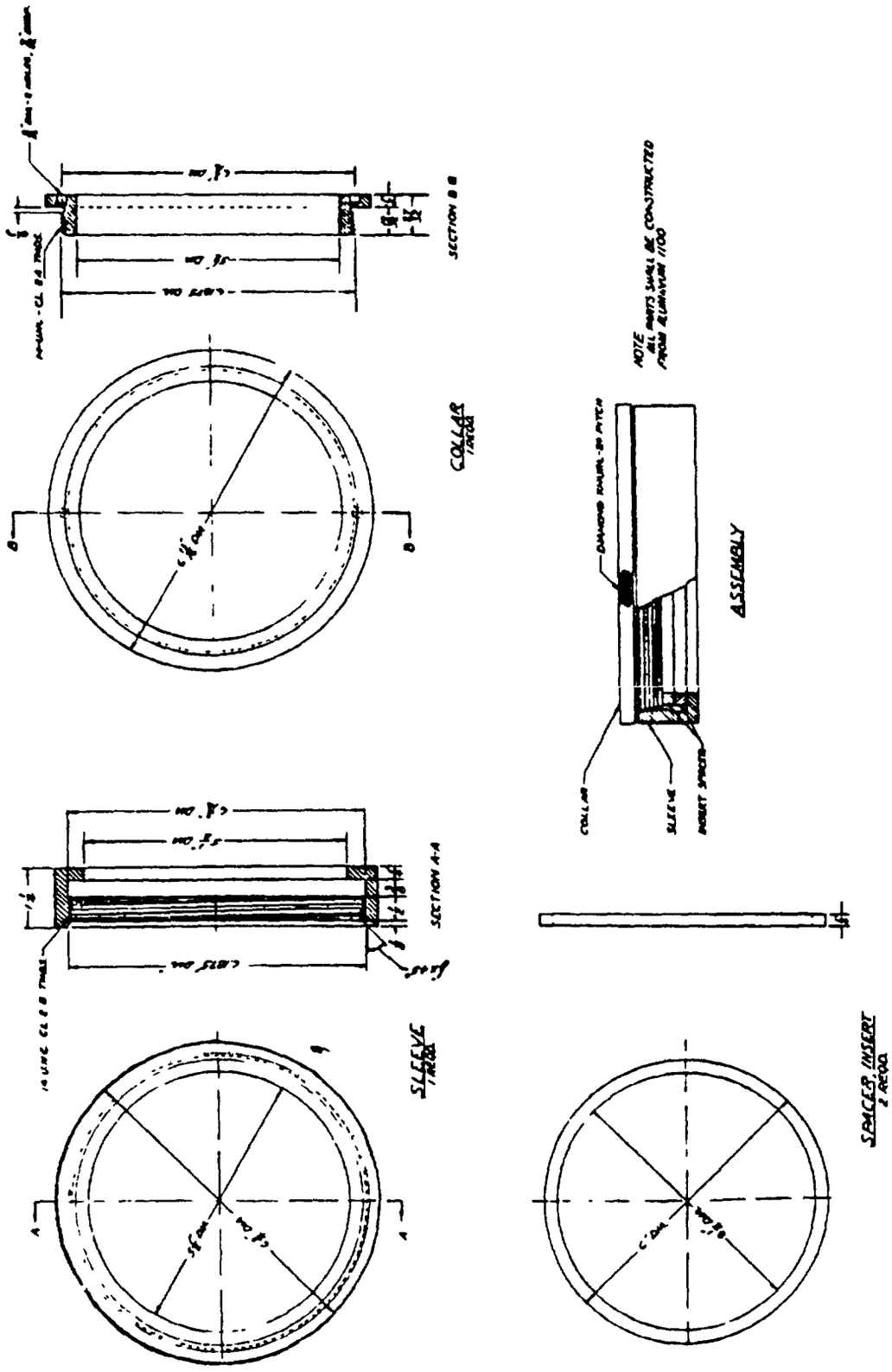


FIGURE 1 Jig, acid testing, no. 2.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T20:28Z
Check the source to verify that this is the current version before use.

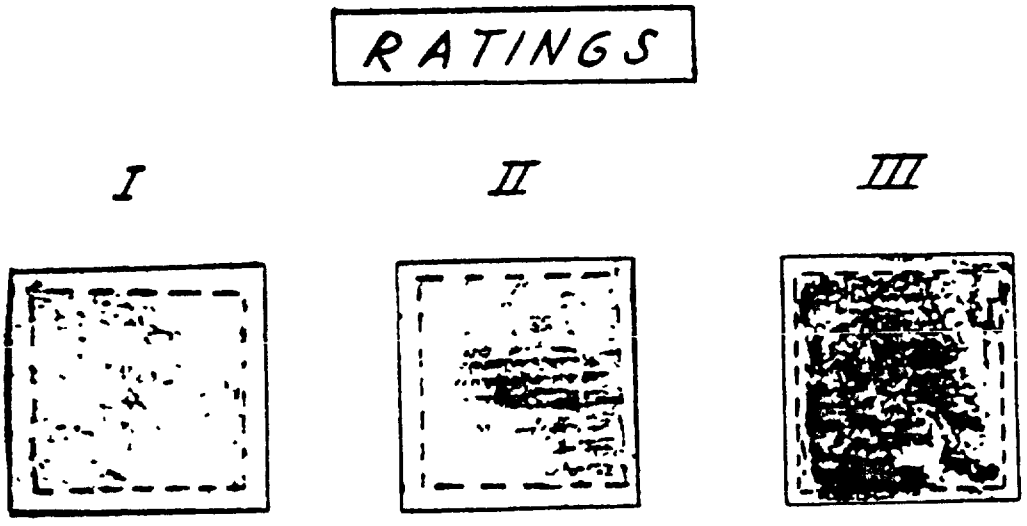


FIGURE 2. Surface tack standards - nitric acid test.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T20:28Z
Check the source to verify that this is the current version before use.

INSTRUCTIONS. In a continuing effort to make our standardization documents better, the DoD provides this form for use in submitting comments and suggestions for improvements. All users of military standardization documents are invited to provide suggestions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and mailed. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, was too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered.

NOTE. This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

(Fold along this line)

(Fold along this line)

DEPARTMENT OF THE ARMY

US Army Natick Research, Development, and
Engineering Center
Natick, MA 01760-5014

NO POSTAGE
NECESSARY
IF MAILED
IN THE
UNITED STATES

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300
STRNC-ES

BUSINESS REPLY MAIL
FIRST CLASS PERMIT NO 12062 WASHINGTON D C
POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE ARMY

Commander
US Army Natick Research, Development,
and Engineering Center
ATTN: STRNC-ES
Box 14A
Natick, MA 01760-5014

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T20:28Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1 DOCUMENT NUMBER	2 DOCUMENT TITLE
MIL-C-43062D	Cloth, Coated, Nylon, Resin Modified Butyl Coated, Acid And Fuel Resistant

3a NAME OF SUBMITTING ORGANIZATION	4 TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)* _____

5 PROBLEM AREAS

a. Paragraph Number and Wording

b. Recommended Wording

c. Reason/Rationale for Recommendation

6 REMARKS

7a NAME OF SUBMITTER *(Last, First, MI)* – Optional	b WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c MAILING ADDRESS *(Street, City, State ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE

NATICK/OF, 1 Apr 84

Source: https://assist.dla.mil - Downloaded: 2026-10-21T20:28Z
Check the source to verify that this is the current version before use.